# Zero-Budget Lab Environment Guide

*Start small, use local sample data, and add SIEM components only when ready*

## Design rule: the smallest useful lab first

A student does not need to install a complete SOC stack on day one. Start with a browser, a text editor, bundled sample events, and the resource directory. Add local logs and virtual machines only if the computer can handle them.

| Tier | Environment | Best for | Estimated cost |
|---|---|---|---|
| Tier 0: no installation | Browser + OpenCyberPath notes + sample JSON/CSV | Foundations, reading logs, triage, metric calculations | ₹0 in software for bundled files; internet/computer still needed |
| Tier 1: local basics | Windows Event Viewer or existing Linux logs + Python | Inspect own authorised logs, parse data, build reports | Can be ₹0 with existing OS and tools |
| Tier 2: local SIEM | A small local Wazuh deployment or suitable VM setup | Agents, ingestion, detection rules, dashboards | Software is open source; hardware and electricity still matter |
| Tier 3: advanced integration | Windows endpoint + Wazuh manager/indexer/dashboard + OpenCyberPath | End-to-end telemetry and alert investigation | Potentially ₹0 in software, but requires adequate resources and setup |

## Basic environment skills to learn

- File paths and permissions in Windows and Linux.
- Processes, services, users, and basic command-line navigation.
- IP address, DNS, ports, HTTP/HTTPS, and local networking.
- JSON and CSV formats; timestamps and time zones.
- Virtual machines, snapshots, networking modes, and safe lab isolation.
- Log sources, event fields, detection rules, and evidence handling.

## Before installing a full SIEM

1. Check the official system requirements for the current version.
2. Check available RAM, CPU, disk space, and whether virtualisation is supported.
3. Create a VM snapshot or back up important files before changing your lab.
4. Keep intentionally vulnerable applications isolated and never expose them publicly.

## Reference setup path

1. Follow the current official Wazuh Quickstart and installation requirements instead of old copied commands.
2. Start with one authorised test endpoint and one manager. Add components only as required and as hardware permits.
3. Generate a harmless test event and confirm every step: endpoint, agent/collector, manager alert, application ingestion, database record, detection, incident, and analyst note.
4. Capture evidence at each stage. A running service does not prove successful delivery of events.
5. Shut down local virtual machines and services when not in use to conserve resources.

## Official docs and learning videos

- Wazuh Getting Started
- Wazuh installation guide
- Splunk free training
- Microsoft Sentinel skill-up training

**COST WARNING** Open-source software is not the same as zero operating cost. Full SIEM labs may need substantial resources. Avoid cloud labs or trials until you know their billing terms.